# Illya Yurchanka

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Physics M.Sc. student at the University of Warsaw with strong programming skills in Java, C++, Python, and SQL. Experience building production-ready software systems and applying data-driven analysis to solve complex scientific problems. Passionate about writing clean, efficient code and delivering end-to-end solutions that empower organizations to make informed decisions. Seeking software engineering roles where I can contribute to mission-critical data platforms.

## SKILLS

**Programming:** Python, Java, C++ (STL, Eigen, Boost), SQL (MySQL, PostgreSQL, DuckDB), JavaScript/TypeScript

**Data Structures & Algorithms:** Trees, Graphs, Hash Tables, Sorting, Searching, Dynamic Programming, Complexity Analysis

**Data Platforms & Engineering:** Spark, Pandas, NumPy, Scikit-Learn, PyTorch, ML Pipelines, ETL, Data Processing

**Tools & DevOps:** Git, Docker, Linux/Bash, CMake, Make, ROOT/TMVA, PyTest, CI/CD

**Languages:** English C1, Polish C1, Russian (Native), Belarusian (Native)

## EDUCATION

**University of Warsaw** | *M.Sc. in Physics* **Oct 2025 – Jun 2027 (expected)**

- GPA: 4.5/5 Focus: Particle Physics & Machine Learning applications

**University of Warsaw** | *B.Sc. in Physics* **Oct 2021 – Jun 2025**

- GPA: 4.5/5 Thesis: Identification of $\Lambda$ hyperons in CBM spectrometer using TMVA (ML)

## EXPERIENCE

**M.Sc. Researcher – CBM Experiment, GSI/FAIR (Darmstadt) Oct 2025 – Present**

- Built and owned full data processing pipeline from ingestion to analysis output; designed and implemented C++ workflows for the CBM detector used in high-energy physics experiments.
- Applied ML techniques to simulated Au+Au collision data to improve particle identification and background rejection; results inform detector commissioning strategy.
- Collaborated with international team of 15+ researchers to integrate analysis modules into the experiment's core software framework.

**Research Intern – General Relativity Mar 2025 – Jun 2025**

Center for Theoretical Physics, Polish Academy of Sciences

- Conducted independent research on general relativity; reviewed 10+ peer-reviewed publications to map the state-of-the-art and identify gaps for novel theoretical contributions.
- Built mathematical models and performed symbolic/numerical computations in Mathematica and SciPy, producing analytical results presented to senior researchers.

**Goldman Sachs Warsaw Hackathon – Quant Challenge Dec 2024**

- Led end-to-end implementation of Monte Carlo simulation framework for financial market forecasting; delivered complete solution from design to final presentation.
- Implemented variance-reduction techniques – bootstrapping and importance sampling – achieving measurably tighter confidence intervals than competing teams.
- Collaborated with team members to integrate quantitative models into unified analysis pipeline under tight deadline.

### Warsaw Econometric Challenge — Apr 2025

- Led a team of 4 in a data science competition; rapidly learned SQL and DuckDB to deliver ML-based predictive solutions under strict deadline.
- Built end-to-end ML pipeline using Scikit-Learn – from data ingestion and feature engineering to model training and predictions.
- Presented results to judges and answered technical questions about methodology and implementation choices.

## PROJECTS

### Multi-Head Latent Attention Implementation — Oct 2025

*Python, PyTorch* [GitHub]

- Implemented MLA mechanism from scratch in PyTorch; engineered efficient KV cache compression through low-dimensional latent space projection.
- Optimized for memory efficiency while preserving attention quality; benchmarked against standard multi-head attention.

### Food Nutrition Analysis with Machine Learning — Jan 2026

*Python, Scikit-Learn, Pandas, NumPy, Transformers* [GitHub]

- Built production-ready ML pipeline on the Open Food Facts dataset: multi-task learning for NutriScore prediction, calorie estimation, NOVA classification, and Eco-Score analysis.
- Implemented semantic search engine using vector embeddings for product recommendation; optimized for retrieval speed.

### Bachelor Thesis – $\Lambda$ Hyperon Identification in CBM Spectrometer — May 2025

*C++, ROOT, ROOT-TMVA*

- Benchmarked KNN, AdaBoost, and MLP classifiers for signal/background separation in high-energy physics data; ensemble methods achieved the best ROC-AUC, directly informing the experiment's offline reconstruction strategy.

### Terminal Cookie Clicker Game — Apr 2024

*C++, Ncurses*

- Built a real-time terminal incremental game with live input handling, persistent game-state management, and a full ncurses TUI rendered from scratch.

## RELEVANT COURSES

**Machine Learning** Neural networks (FC, CNN, GAN, VAE), RNNs, Transformers, NumPy/Pandas/Matplotlib 2023/2024

**Programming & Numerical Methods** Advanced C++, physical simulations, numerical methods in Python 2023/2024

**Statistical Analysis of Experimental Data** Statistical inference, hypothesis testing, data analysis pipelines 2025/2026

**Programming** STL algorithms, templates, containers 2022/2023